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AVAREBIOTECH

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Field Case Study Report of "Avare Biotech" app powered by MAKSA Technology on Frozen-Thawed and Fresh Bovine Semen (Zaria, Nigeria, 2025)

Date: 4th December 2025
Location: National Animal Production Research Institute, A.B.U Zaria, Kaduna, Nigeria.

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Assisted Reproductive Technology Specialist (Kaduna, Nigeria)	**CEO & Co-Founder of Avare Biotech**
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1. Executive Summary / Conclusions

- MAKSA smartphone-based semen evaluation demonstrated consistent agreement with manual hemocytometer counts for concentration and total motility under field and laboratory conditions. The results obtained confirm that mobile semen assessment can provide reliable measurement of core parameters when standard microscope and phone equipment are used
- Progressive motility was recorded by MAKSA; however, manual progressive values were not consistently documented in this experiment, therefore conclusions on accuracy of progressive motility require additional testing.
- Based on these findings, MAKSA can be considered ready for routine field use in bovine semen evaluation, with further validation recommended across additional sample sizes, species, and manual verification protocols.

2. Objective

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for frozen-thawed and fresh bovine semen evaluation against the manual method.

3. Materials and Methods

- Sample Type: Frozen–thawed and fresh bovine semen
- Dilution rate: 0 for the frozen thawed semen, 5 and 10 for the fresh semen
- Devices: Microscope (200x) + Smartphone (iPhone 13)
- Comparison Tools: Manual (hemocytometer) and MAKSA technology
- Software: Avare Biotech App powered by MAKSA technology
- Number of Videos: between 1 and 3 per sample

Table 1 MAKSA results comparison

Bull's Name	Key Semen Parameters	MAKSA	Manual Count	MAKSA vs. Manual Difference	Difference %
Shika	Concentration (M/mL)	27.20	25.40	1.8	7
	Total Motility (%)	99	90	9	9
	Progressive Motility (%)	60			

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Adams	Concentration (M/mL)	28.80	30.60	1.8	6
	Total Motility (%)	90	90	0	0
	Progressive Motility (%)	66			
Isah 1	Concentration (M/mL)	136	150	14	10
	Total Motility (%)	99	90	9	9
	Progressive Motility (%)	81			
Isah 2	Concentration (M/mL)	28.80	30	1.20	4
	Total Motility (%)	99	90	9	9
	Progressive Motility (%)	71			

4. Interpretation

- Motility and concentration match manual method. This confirms that MAKSA's motion-tracking engine is ready for routine use on livestock farms, breeding centers, research institutes and AI technicians.
- Dilution factor of 10 is suitable for fresh bovine semen.
- User-friendly workflow. The test reaffirmed that even with simple equipment, farmers can get reliable motility readings without CASA hardware.
- Reduced time consumption as with the manual method, the MAKSA processed the result within 1 min.

5. Recommendations for Avare

- **Wide Specie Range –** include other animal species like Rabbits and Poultry, research institutes and Universities will benefit from it.
- **Documentation and Comparisons –** make room for cloud storage of Animal specific test conducted at different time, so that different test of the same Animal can be compared.
- **In App Camera** – Make available a camera in the app specific for the app, with that videos can be recorded and uploaded automatically.
- **Error Sign** – When a video is not process successfully, instead of the error sign why not give a suggestion on what is wrong and what can be done, example; increase the dilution factor, or make a clearer video.

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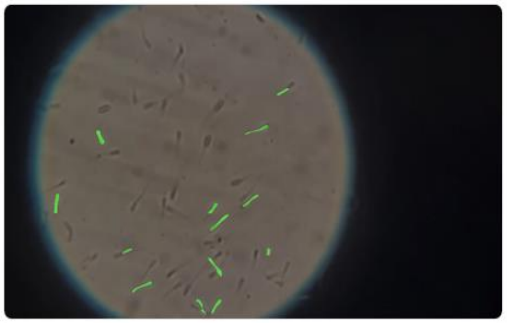
Appendix

Screenshot report samples

Shika

Analysis 01.12.2025

Animal: — **Name:** Shika **Age:** 2 **Breed:** Friesian Cross **Status:** Ready **Created:** 2025-12-01 10:46



Total objects:	17
Concentration (millions/ml^3):	27.20
Dilution rate:	[1:0]
Volume of field of view (mm^3):	0.002775
Progressive Motility:	60%
Non-Progressive Motility:	39%
Immotility:	1%

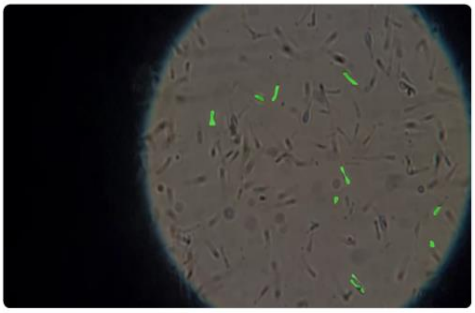
Agglutination

Found:	No

Adams

Analysis 02.12.2025

Animal: — **Name:** Adams **Age:** 5 **Breed:** Friesian Cross **Status:** Ready **Created:** 2025-12-02 10:18



Total objects:	18
Concentration (millions/ml^3):	28.80
Dilution rate:	[1:10]
Volume of field of view (mm^3):	0.001855
Progressive Motility:	66%
Non-Progressive Motility:	37%
Immotility:	1%

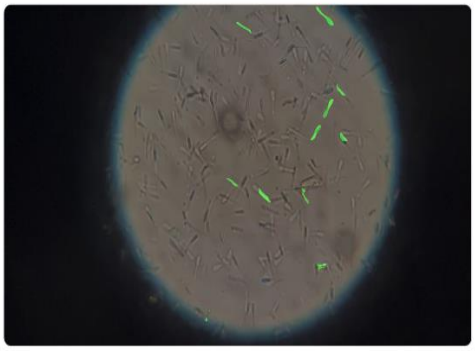
Agglutination

Found:	No

Isah 1

Analysis 03.12.2025

Animal: — **Name:** Isah **Age:** 4 **Breed:** FX **Status:** Ready **Created:** 2025-12-03 09:51



Total objects:	85
Concentration (millions/ml^3):	136
Dilution rate:	[1:10]
Volume of field of view (mm^3):	0.002905
Progressive Motility:	81%
Non-Progressive Motility:	46%
Immotility:	1%

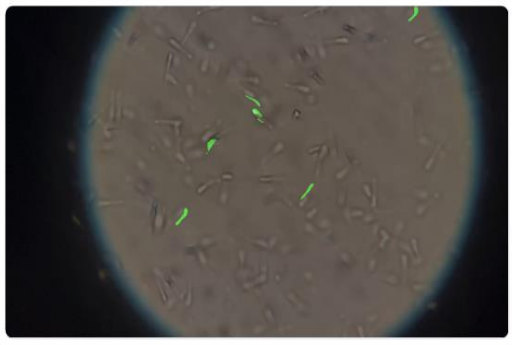
Agglutination

Found:	No

Isah 2

Analysis 03.12.2025

Animal: — **Name:** Isah 2 **Age:** 4 **Breed:** FX **Status:** Ready **Created:** 2025-12-03 09:54



Total objects:	18
Concentration (millions/ml^3):	28.80
Dilution rate:	[1:10]
Volume of field of view (mm^3):	0.001832
Progressive Motility:	71%
Non-Progressive Motility:	51%
Immotility:	1%

Agglutination

Found:	No